

Move grayed out components to right side of the screen

Menus and sub-menus are shown in Ideum Table

Results are shown in the wall screens

Datacenter Sub-menu

Techno-economics

Datacenter Configurator

Advanced Analysis

Graphic showing Technical use cases

2. Storyline for high-fidelity analysis

1. Storyline for Datacenter Configurator:

1. Show datacenter configurator – all the inputs that can be varied and all the assets that can be used
2. Show how two of the available 12 cases result in two different gen. mix builds – Case 1 with CFE of 49% & availability of >3-9s vs. case 2 with CFE of 70% & availability of >5-9s results in higher installed capacity
3. Future roadmap for the configurator – train with more data so that inputs can be continuously varied

P.S. – Given the limited datasets to train, best option would be to not allow a user to change the inputs (load, availability, and CFE) continuously but allow them to toggle between the available datapoints that automatically sets these inputs to some fixed value.

2. Storyline for high-fidelity analysis

Advanced analysis sub-menu

Reference Power Block Architectures (Islanded or Grid Connected)

Advanced Controls

Controller Hardware-in-the-Loop Testing

A B C

A

Ref. Power Block Architecture sub-menu ()

200 MW architecture

Dispatchable wind powered 200 MW architecture

500 MW architecture

D E

B

Advance control concept sub-menu

HD + LM power sharing control

Renewable powered datacenters

F

C

CHIL sub-menu

Test Ref. Architecture Operation

Test Advanced Controls Performance

G

Only 200 MW architecture ready for now

D

Ideum Screen

200 MW arch. Sub-menu

Baseline Configuration

LM2500 Powered Island

Damped vs Undamped as toggle switches?

Augmented Configurations

Add BESS

Add GFMe

Add GFMe + BESS

Add MVUPS

Add SST

Use Cases

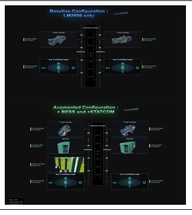
Forced Power Oscillation

Contingency scenarios

SSTI

Ride through performance

Wall screens



Tech. Specs. for Baseline
- 200 MW islanded system
- 8 LM2500 rated @ 35MW
- 4 Datacenter Load centers
- x MW IT Load, y MW Non-IT Load, PUE = 1.z
- Connected at 34.5 kV MVAC bus

Tech. Specs. for Augmented
- Baseline specs +
- 2 GFM BESS plants rated @ x MW/y MWh each

E

Ideum Screen

Dispatchable wind powered 200 MW arch. Sub-menu

Baseline Configuration

200 MW GFL Wind + 200 MW GFMe BESS

Augmented Configurations

Make wind GFM

Add Stability Solutions

Add Solar

Hyperscaler Architecture

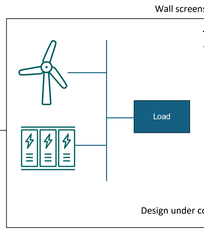
Use Cases

Change wind speed

Torsional oscillations

LVRT

Wall screens



Tech. Specs. for Baseline
- TBD

Tech. Specs. for Augmented
- TBD

Design under construction

G

Ideum Screen

Test Ref. Architecture sub-menu

Baseline Configuration

1 MJOs + (disconnected) + 1 Load + 1 Load (disconnected)

Augmented Configurations

Connect 2nd load

Connect LM2500

Add multiple LM2500

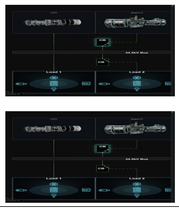
Add EMS

Use Cases

Trip LM

Trip Load

Wall screens



Tech. Specs. for Baseline
- HD rating
- LM rating
- Load rating

Tech. Specs. for Augmented
- TBD

F

Ideum Screen

Test Ref. Architecture sub-menu

Baseline Configuration

8 LM2500s + 2 BESS, 200 MW system

Augmented Configurations

Add MVUPS + SST

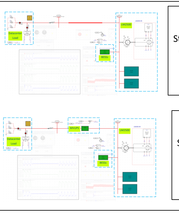
Add SST & Storage

Use Cases

Baseload

Al Load

Wall screens



Streams data for load in MW and one of the generator power in MW for baseline system

Streams data for load in MW and one of the generator power in MW for augmented system

Need to verify the feasibility of this with Bruce & GA team